

Module 1 Chapter 3

Conditional Statements and Truth Tables

Sometimes (quite often!) we want to express statements of the form “**If P then Q**” that is “**If P is True then Q is also True**”. Such expressions can also be written as “**P implies Q**”.

In other words, we can say that “**Whenever P is True, Q is also true**”. Such statements are called **Conditional Statements**. Here, P is called the **Hypothesis** while Q is called the **Conclusion**.

The notation used for Implies is “ $\Rightarrow$ ” or “ $\rightarrow$ ”. So, P implies Q can be written as

$$\underset{\text{(hypothesis)}}{P} \Rightarrow \underset{\text{(conclusion)}}{Q} \text{ or } P \rightarrow Q.$$

When can we say that $P \Rightarrow Q$ is True?

Well, when P is True and Q is True, $P \Rightarrow Q$ is true (From the definition)

When P is True and Q is False, we know $P \Rightarrow Q$ is False.

What about when P is False?

The definition of an implies statement doesn't say anything about the case when P is False, so, when P is False, no matter what the truth value of Q is, $P \Rightarrow Q$ is True.

$P \rightarrow$ working hard
 $Q \rightarrow$ getting A

$P \rightarrow Q$

$P \rightarrow T \rightarrow P \rightarrow Q \rightarrow \text{True}$
 $Q \rightarrow T$

Let's take an example to understand better:

Let's say that I make the claim *"If you work hard, then you will get an A in this course"*.

$\left. \begin{matrix} P \rightarrow T \\ Q \rightarrow T \end{matrix} \right\} P \rightarrow Q \rightarrow \text{True}$
 $\left. \begin{matrix} P \rightarrow F \\ Q \rightarrow F \end{matrix} \right\} P \rightarrow Q \rightarrow \text{True}$
 $\left. \begin{matrix} P \rightarrow F \\ Q \rightarrow T \end{matrix} \right\} P \rightarrow Q \rightarrow \text{True}$

Here, P is "You work hard" and Q is "You will get an A". When can you call me a liar? When can you declare I told the truth? Certainly, if you work hard and get an A, I told the truth. Also, if you work hard and don't get an A, then I lied to you.

$\left. \begin{matrix} P \rightarrow T \\ Q \rightarrow F \end{matrix} \right\}$
 $P \rightarrow Q : \text{false}$

However, if you don't work hard, then no matter what happens, you don't get to call me a liar! My claim didn't cover your situation; I was assuming all along you would just work hard! Thus, I didn't speak an untruth and so, I did speak the truth.

Such a situation where P is False, so, $P \Rightarrow Q$ is True, is called **"False Hypothesis"**.

When the statement on the left of the " $\Rightarrow$ " does not hold, then we are not in a situation that is meant to be addressed by the claim, so we cannot assert that the claim is False. Therefore, the claim must be True .

Let's Practice

Determine the truth values of the following mathematical statements:

- $\overset{T}{(1+2=3)} \Rightarrow (\forall x \in \mathbb{R}. x^2 \geq 0) \overset{T}{\text{True}}$
- $\overset{F}{(1+2=5)} \Rightarrow (\forall x \in \mathbb{R}. x^2 \geq 0) \overset{T}{\text{True}} \quad F \cdot H$
- $\overset{F}{(1+3=5)} \Rightarrow (\text{Donald Trump is our new president}) \overset{T}{\text{True}}$
- $\overset{T}{(1+6=7)} \Rightarrow (\overset{F}{0=2}) \overset{F}{\text{False}}$
- $\overset{T}{(1=1)} \Rightarrow (\exists x \in \mathbb{R}. x^2 < 0) \overset{F}{\text{False}}$
- $\overset{T}{(\text{Pythagorean Theorem})} \Rightarrow (\overset{T}{2=2}) \overset{T}{\text{True}}$
- $\overset{F}{(0=1)} \Rightarrow (\overset{F}{3=7}) \overset{T}{\text{True}}$

Truth Tables

Let's look at an example:

$\begin{matrix} P \\ T \\ T \\ F \\ F \\ F \\ F \end{matrix}$
 $\begin{matrix} Q \\ T \\ F \\ F \\ F \\ F \\ F \end{matrix}$
 $\begin{matrix} R \\ T \\ F \\ T \\ F \\ F \\ F \end{matrix}$

2^2

2^3

2^n

$n = \# \text{ of ind. claims}$

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$Q \Rightarrow P$
T	T	F	T	T	T	T
T	F	F	F	T	F	T
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Fig 1

Here's the main idea:

Columns: Mathematical Statements.

Values in the columns: Corresponding Truth Values of the mathematical Statements.

Rows: Assignment of truth values to the constituent statements (P and Q here).

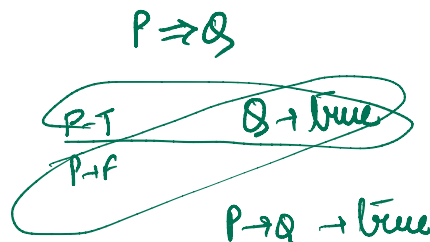
We will find uses of Truth tables especially for checking claims/ proofs of claims as we will see in the later sections!

Something Interesting:



Let's revisit $P \Rightarrow Q$

$\neg P \rightarrow \text{true}$
 $Q \rightarrow \checkmark$
 $Q \rightarrow \text{true}$



If P is False, the entire statement is True, but if P is True, we need Q to be True to declare the entire statement to be True.

$$P \Rightarrow Q \quad \neg P \vee Q$$

We can, thus, say that there are two ways in which a conditional statement can hold and write these two ways using an “OR” statement:

Either $\neg P$ holds (i.e. a false hypothesis), or else Q holds. In either of these situations, the conditional statement $P \Rightarrow Q$ must hold!

More formally, we can say:

The conditional statement " $P \Rightarrow Q$ " and the statement " $\neg P \vee Q$ " have the same truth value.

logically equivalent

Can we check this claim using Truth Tables?

P	Q	$\neg P$	$\neg P \vee Q$	$P \Rightarrow Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Fig 2



Let's Practice:

Let S and T be arbitrary sets. Let P be " $S \subseteq T$ ", and let Q be " $S - T = \emptyset$ ".

Find the truth value of:



$$S = \{1, 2, 3\}$$

$$T = \{a, b, c\}$$

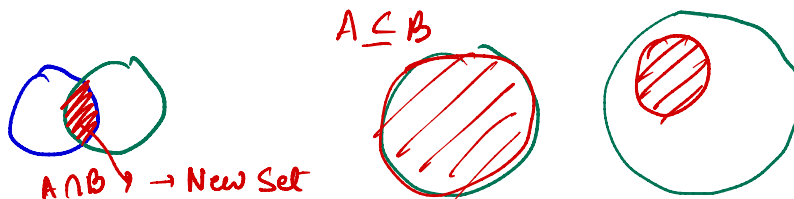


$$T - S = \{a, b\}$$

- $P \Rightarrow Q$ True
- $Q \Rightarrow P$ True
- $\neg Q \Rightarrow \neg P$ True
- $\neg P \Rightarrow \neg Q$ True

Sets and Logical Connectives:

Subsets:



We write $A \subseteq B$ if the following holds: whenever $x \in A$, we can also say $x \in B$.

Notice the key word, “whenever”; this indicates that we need both a universal quantification and a conditional statement.

• → |

Definition: Let A, B, U be sets, where $A, B \subseteq U$ (i.e. U is a universal set).

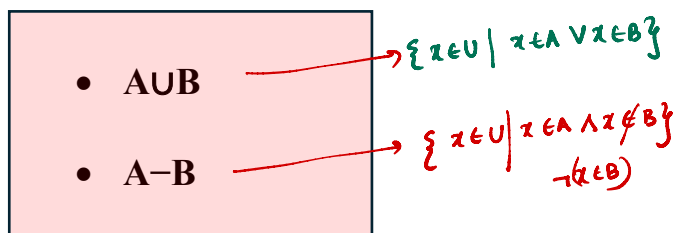
$A \subseteq B$ is true if:

$$\forall x \in U. x \in A \Rightarrow x \in B$$

Let's try to define other set operations:

$$A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$$

Try:



$A \subseteq B$
all elements of A are in B
If an element is in A , it is in B
 $\Rightarrow$
 $\rightarrow$
An element in A implies that element is in B
 $A \cap B$
 $\rightarrow$ New set :
element belongs to A AND
" " " B
 $\wedge$
{ }

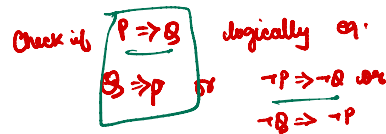
Converse, Inverse and Contrapositive

Let P and Q be mathematical statements. Consider the “original” claim, $P \Rightarrow Q$.

Converse: $Q \Rightarrow P$

Inverse: $\neg P \Rightarrow \neg Q$ *NOT A NEGATION*

Contrapositive: $\neg Q \Rightarrow \neg P$



Let's Practice:

Write the converse, inverse and contrapositives of the following statements:

- If two angles are congruent, they have the same measure
- An acute angle is less than 90°
- $x > 4$ implies $x + 3 > 7$

We know that converses may not have the same truth value as the original statement. What about contrapositives? Use Truth Tables to find out!

Logical Equivalence

If P and Q ~~are~~ ^{are} two given statements, we say P is logically equivalent to Q if P and Q have the same truth value regardless of whether it's T or F .

We use the symbol " $\Leftrightarrow$ " to represent logical equivalence. They are also called as "If and only if" statements or "Bi-conditional" statements.

Let's look at the Truth Table:

P	Q	$P \Rightarrow Q$	$\neg P$	$\neg Q$	$\neg Q \Rightarrow \neg P$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T



P	Q	$\neg P$	$\neg P \vee Q$	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \Leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	F	T	F
F	T	T	T	T	F	F
F	F	T	T	T	T	T

Fig 3

$$P \Leftrightarrow Q$$

$$P \Rightarrow Q \wedge$$

$$Q \Rightarrow P$$

Note that $P \Leftrightarrow Q$ has truth value T if and only if

$P \Rightarrow Q \wedge Q \Rightarrow P$ is true.

So, we can say in other words that P is logically equivalent to Q only when P implies Q is True and Q implies P is True

We can also see from the above table that

$$P \Rightarrow Q \Leftrightarrow \neg P \vee Q$$

Let's Practice:

Prove the following using Truth Tables:

- $\neg (P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$
- $\neg (P \Rightarrow Q) \Leftrightarrow P \wedge \neg Q$

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

$\therefore \neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$

P	Q	$P \Rightarrow Q$	$\neg(P \Rightarrow Q)$	$\neg Q$	$P \wedge \neg Q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

$\therefore \neg(P \Rightarrow Q) \Leftrightarrow P \wedge \neg Q$

Biconditionals are also used in definitions to define the property of a particular term. For example:

We can define an *even integers* as:

$$x \in \mathbb{Z} \text{ is even} \Leftrightarrow \exists k \in \mathbb{Z}. x = 2k$$

Can you define odd integers?

1. Given,

$$P : x + 7 = 11$$

$$Q : x = 5$$

Problem : Is the following True?
 $P \Leftrightarrow Q$

2. Is the following biconditional statement true:
 $x = 3$ if and only if $x^2 = 9$

Given,

$$P : x + 7 = 11$$

$$Q : x = 5$$

Problem : Is the following True?

$$P \Leftrightarrow Q$$

$P \Rightarrow Q$ represents "If $x + 7 = 11$ then $x = 5$ "

$Q \Rightarrow P$ represents "If $x = 5$ then $x + 7 = 11$ "

$P \Rightarrow Q$ is false : when P is true, Q is false

$Q \Rightarrow P$ is false : when Q is true, P is false

$$(P \Leftrightarrow Q) \Leftrightarrow (P \Rightarrow Q \wedge Q \Rightarrow P)$$

$\therefore (P \Rightarrow Q \wedge Q \Rightarrow P)$ is false
 $\therefore P \Leftrightarrow Q$ is false

OR

when P is true, Q is not true

when Q is true, P is not true

$\therefore P$ and Q do not have the same truth value $\therefore P \Leftrightarrow Q$ is False

Is the following biconditional statement true:
 $x = 3$ if and only if $x^2 = 9$

Let P be " $x = 3$ "

Let Q be " $x^2 = 9$ "

$P \Rightarrow Q$ is true i.e. whenever P is true,
 Q is true

$Q \Rightarrow P$ is false i.e. whenever Q is true,
 P is not true

Eg. when $x = -3$

Q is true but P is not true

$\therefore P \Leftrightarrow Q$ is false